

RStudio Series02

Data Mining and Statistical Tests

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Data mining and statistical tests

Data mining is the process of exploring data to discover useful patterns and relationships.

Objective

The objective is to explore, visualize, and statistically analyze different datasets using R. The analysis includes descriptive statistics, hypothesis testing, group comparisons, correlations, and an introduction to more advanced data-analysis methods.

A. Clinical Problem

RQ: Is proximal arm non-use (**PANU**) related to shoulder non-use (**SANU**) and or elbow non-use (**EENU**) after stroke?

Variables

After a stroke, a patient may compensate for reduced shoulder or elbow use by leaning the trunk forward more. Therefore, we want to know whether:

- more **shoulder non-use** is associated with more **proximal arm non-use**
- more **elbow non-use** is associated with more **proximal arm non-use**

This can be studied using the **correlations between PANU, SANU and EENU**.

Variable	Description	Unit	Range
SANU	shoulder flexion the patient can use with the trunk restrained but does not use spontaneously.	%	0-100
EENU	elbow extension the patient can use with the trunk restrained but does not use spontaneously.	%	0-100
PANU	shoulder and elbow movement together that the patient can use with the trunk restrained but does not use spontaneously.	%	0-100

+ = non-use : patient used more movement in the restrained condition than in the spontaneous condition

- = over-use : patient used more movement spontaneously than in the restrained condition

0 → no difference between the two conditions.

$PANU (\%) = (\text{hand displacement} - \text{trunk displacement}) / \text{hand displacement}$

B. Experimental Methodology

Participants with a previous stroke performed a seated reaching task with the affected arm while their movement was recorded using motion capture.

Condition 1: Spontaneous

- The participant reaches for the cone using their natural movement strategy.
- Trunk compensation is allowed.
- This shows **how the participant normally chooses to perform the movement**.

Condition 2: Trunk Restrained

- The participant performs the same reach while keeping the trunk still.
- This reduces forward trunk compensation.
- The participant must rely more on shoulder and elbow movement.
- This shows **what the affected arm can do when trunk compensation is restricted**.

C. Data: How Non-Use Is Identified

- If the participant produces more shoulder and or elbow movement when the trunk is restrained than during spontaneous reaching this shows that they were **capable of using that movement but did not use it spontaneously**.
- **Non-use** = movement capacity available – movement used spontaneously

Variable	Correlation Predicition	Why?	Unit	Range
SANU	Positive	More shoulder non-use should mean more overall proximal-arm non-use.	%	0-100
EENU	Positive	More elbow non-use should mean more overall proximal-arm non-use.	%	0-100
PANU	Positive	Both contribute to reaching, but one joint can partly compensate for the other.	%	0-100

1. Import data

```
# 1. Import libraries
library(tidyverse) # tools for plotting and data analysis
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0 —
✓ dplyr      1.2.1    ✓ readr      2.2.0
✓ forcats    1.0.1    ✓ stringr    1.6.0
✓ ggplot2    4.0.3    ✓ tibble     3.3.1
✓ lubridate  1.9.5    ✓ tidyr      1.3.2
✓ purrr      1.2.2

— Conflicts — tidyverse_conflicts() —
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
# 2. Import the dataset
nonuse <- read_csv("data/NonUse.csv")
```

```
Rows: 432 Columns: 3
```

```
— Column specification —————
```

```
Delimiter: ","
```

```
dbl (3): PANU, EENU, SANU
```

```
i Use `spec()` to retrieve the full column specification for this data.
```

```
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# View the first 6 rows to make sure it was imported correctly
```

```
head(nonuse)
```

```
# A tibble: 6 × 3
```

```
  PANU  EENU  SANU  
  <dbl> <dbl> <dbl>  
1 -1.92   5.36 -4.84  
2  1.12  -2.51  1.74  
3  1.84   5.95 -1.44  
4  0.0706 4.04 -1.67  
5  3.15   NA   NA  
6  1.86   NA   NA
```

```
# summary of the data
```

```
summary(nonuse)
```

PANU		EENU		SANU	
Min.	:-3.306	Min.	:-33.9780	Min.	:-38.4178
1st Qu.:	1.629	1st Qu.:	-0.4864	1st Qu.:	-3.4968
Median :	4.370	Median :	6.2732	Median :	0.9708
Mean :	9.742	Mean :	8.3701	Mean :	1.3908
3rd Qu.:	13.919	3rd Qu.:	15.3383	3rd Qu.:	7.0470
Max.	:68.286	Max.	:50.1748	Max.	:40.1735
NAs	:13	NAs	:223	NAs	:223

2. Check the imported data

```
str(nonuse) # structure of data
```

```
spec_tbl_ [432 × 3] (S3: spec_tbl_df/tbl_df/tbl/data.frame)  
 $ PANU: num [1:432] -1.9181 1.1229 1.8406 0.0706 3.1523 ...  
 $ EENU: num [1:432] 5.36 -2.51 5.95 4.04 NA ...  
 $ SANU: num [1:432] -4.84 1.74 -1.44 -1.67 NA ...  
 - attr(*, "spec")=  
 .. cols(  
 ..   PANU = col_double(),  
 ..   EENU = col_double(),  
 ..   SANU = col_double()  
 .. )  
 - attr(*, "problems")=<pointer: 0xbc08338b0>
```

3. Clean data

The data set contains missing values, only complete pairs of observations were used for each analysis. Thus the data must be cleaned before performing any correlation analysis.

```
# Keep only complete observations for all three variables
nonuse_clean <- nonuse %>%
  select(PANU, SANU, EENU) %>%
  drop_na() # removes rows with NA
```

```
head(nonuse_clean) # check the new data table
```

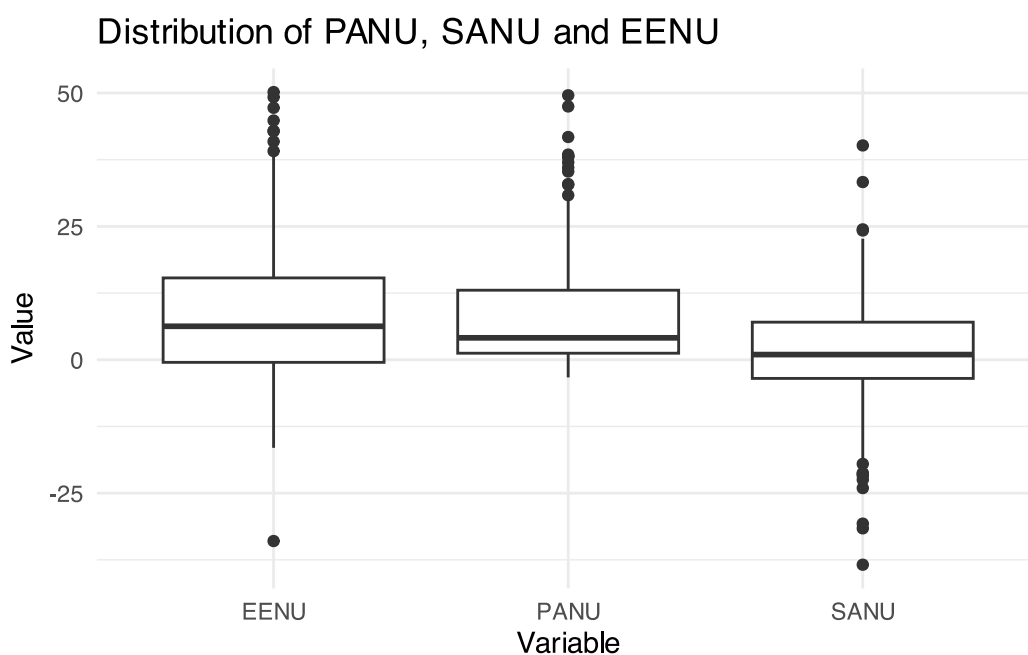
```
# A tibble: 6 × 3
  PANU  SANU  EENU
<dbl> <dbl> <dbl>
1 -1.92  -4.84  5.36
2  1.12   1.74 -2.51
3  1.84  -1.44  5.95
4  0.0706 -1.67  4.04
5 -1.10 -10.3  16.4
6  1.77  -3.56  4.21
```

4. Distribution of data

Before the correlation analysis, boxplots were used to inspect the distribution, spread, and possible outliers of PANU, SANU, and EENU.

```
# 1. Convert the data from wide to long format
nonuse_long <- nonuse_clean %>%
  pivot_longer(cols = c(PANU, SANU, EENU), names_to = "Variable", values_to = "Value")
# convert the data into a two columns one with variable names and the other with values because
# ggplot requires this data formatting

# 2. Create one boxplot for all three variables
ggplot(nonuse_long, aes(x = Variable, y = Value)) +
  geom_boxplot() +
  labs(title = "Distribution of PANU, SANU and EENU", x = "Variable", y = "Value") +
  theme_minimal()
```



5. Correlation analysis

- Visualize PANU–SANU and PANU–EENU
- Calculate the correlations
- Interpret the correlation coefficients
- Perform the two linear regressions
- Extract the regression equations
- Write a scientific-results sentence
- Discuss limitations

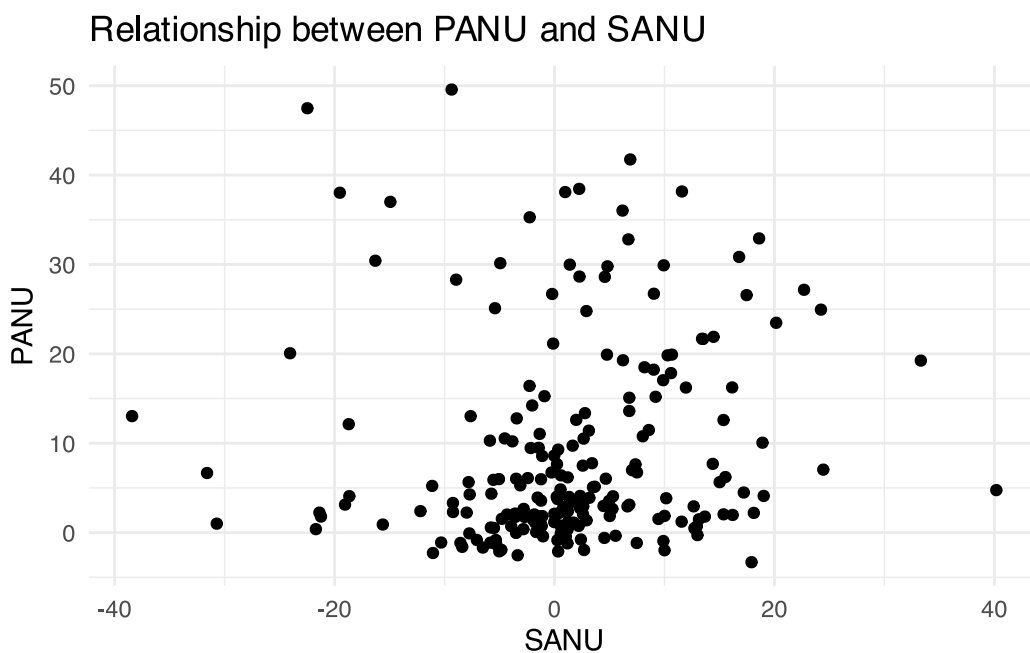
a). Visualize relationships: scatterplot of PANU-SANU and PANU-EENU

Q: As SANU or EENU changes, does PANU tend to change as well?

- SANU or EENU as the **x-variable**
- PANU as the **y-variable**

```
# 1. Create a scatterplot for PANU-SANU
```

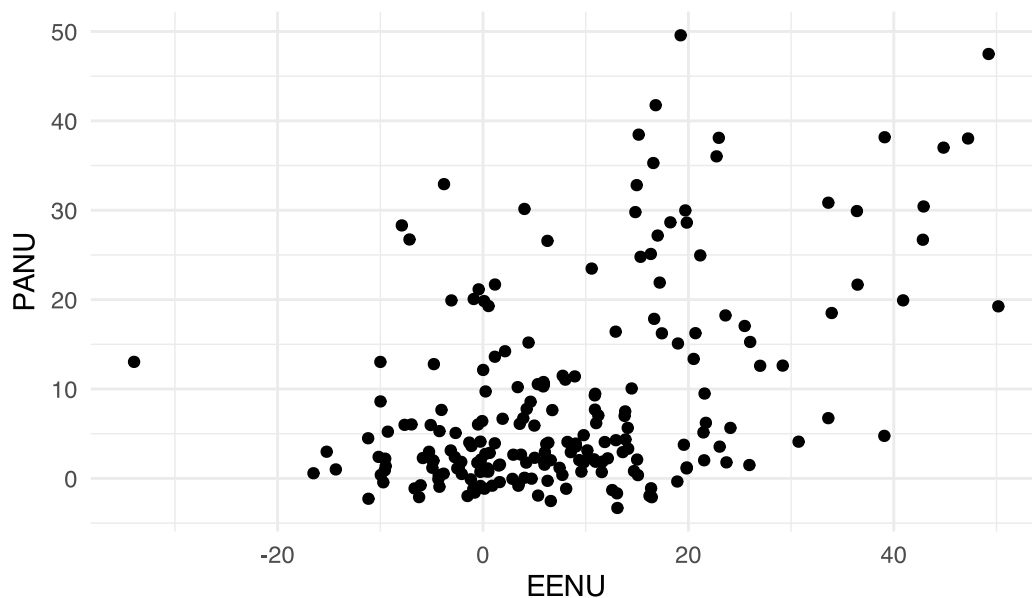
```
ggplot(nonuse_clean, aes(x = SANU, y = PANU)) +  
  geom_point() + # represent each observation as a point  
  labs(title = "Relationship between PANU and SANU", x = "SANU", y = "PANU") +  
  theme_minimal()
```



```
# 2. Create a scatterplot for PANU-EENU
```

```
ggplot(nonuse_clean, aes(x = EENU, y = PANU)) +  
  geom_point() +  
  labs(title = "Relationship between PANU and EENU", x = "EENU", y = "PANU") +  
  theme_minimal()
```

Relationship between PANU and EENU



Interpretation

The PANU–EENU relationship appears more clearly positive, while the PANU–SANU relationship shows greater dispersion. Several observations are far from the main cluster, which may influence the correlation. The strength of these relationships will be quantified using correlation coefficients.

b). Linear regression: PANU-SANU and PANU-EENU

General form: $y = ax + b$

Metrics analyzed:

1. $p < 0.05 \rightarrow$ statistically significant
2. R^2 = proportion of variation explained by the linear model
 - $R^2 \rightarrow (0 \text{ to } 1)$
3. r = direction + strength of linear association
 - $r \rightarrow (-1 \text{ to } +1)$

```
# 1. Linear regression: PANU-SANU
```

```
model_SANU <- lm(PANU ~ SANU, data = nonuse_clean) # store the linear model lm().
# PANU = left side of equation (variable we want to predict), SANU = right side of equation
# (predictor variable)
```

```
summary(model_SANU) # show the main regression results (intercept b, slope a)
```

Call:

```
lm(formula = PANU ~ SANU, data = nonuse_clean)
```

Residuals:

Min	1Q	Median	3Q	Max
-13.983	-7.784	-4.734	4.933	41.648

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	8.86819	0.78242	11.334	<2e-16 ***
SANU	0.10093	0.07355	1.372	0.171

```

---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 11.21 on 207 degrees of freedom
Multiple R-squared:  0.009016, Adjusted R-squared:  0.004229
F-statistic: 1.883 on 1 and 207 DF,  p-value: 0.1714

```

```

# 2. Linear regression: PANU-EENU

model_SANU <- lm(PANU ~ EENU, data = nonuse_clean)

summary(model_SANU)

```

```

Call:
lm(formula = PANU ~ EENU, data = nonuse_clean)

Residuals:
    Min       1Q   Median       3Q      Max
-16.487  -6.555  -2.321   3.249  36.230

Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  5.67295    0.80975   7.006 3.39e-11 ***
EENU         0.39851    0.05141   7.752 4.00e-13 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 9.917 on 207 degrees of freedom
Multiple R-squared:  0.225, Adjusted R-squared:  0.2213
F-statistic: 60.1 on 1 and 207 DF,  p-value: 4.005e-13

```

PANU-SANU

Regression equation

$$\text{PANU} = 0.10093 * \text{SANU} + 8.86819$$

Slope (a) = 0.10093 → positive relationship. Intercept (b) = 8.86819.

$$p = 0.171$$

0.171 > 0.05, so the slope is not statistically significant. Although the slope is positive, the data do not provide sufficient evidence of a linear relationship between PANU and SANU.

$$R^2 = 0.009016$$

The regression shows a very weak relationship between SANU and PANU. SANU explains approximately 0.9% of the variation in PANU.

PANU-EENU

Regression equation

$$\text{PANU} = 0.39851 * \text{EENU} + 5.67295$$

Slope (a) = 0.39851 → positive relationship. Intercept (b) = 5.67295.

$$p = 4.005e-13$$

4.005e-13 < 0.05, so the slope is statistically significant. The data provide sufficient evidence of a linear relationship between PANU and EENU.

$$R^2 = 0.225$$

The regression shows a stronger relationship between EENU and PANU. EENU explains approximately 22.5% of the variation in PANU.

c). Pearson's r correlation test: PANU-SANU and PANU-EENU

```
# 1. Pearson correlation between PANU and SANU
```

```
cor.test(nonuse_clean$PANU, nonuse_clean$SANU, method = "pearson")
```

Pearson's product-moment correlation

```
data: nonuse_clean$PANU and nonuse_clean$SANU
t = 1.3723, df = 207, p-value = 0.1714
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 -0.04129318  0.22773346
sample estimates:
      cor
0.09495356
```

```
# 2. Pearson correlation between PANU and EENU
```

```
cor.test(nonuse_clean$PANU, nonuse_clean$EENU, method = "pearson")
```

Pearson's product-moment correlation

```
data: nonuse_clean$PANU and nonuse_clean$EENU
t = 7.7524, df = 207, p-value = 4.005e-13
alternative hypothesis: true correlation is not equal to 0
95 percent confidence interval:
 0.3619381 0.5731684
sample estimates:
      cor
0.4743524
```

PANU-SANU

r = 0.09495356

The correlation coefficient is very close to 0, indicating a very weak positive linear correlation between SANU and PANU.

PANU-EENU

r = 0.4743524

The correlation coefficient indicates a moderate positive linear correlation between EENU and PANU.

d). Result Interpretation

PANU was not significantly associated with SANU ($r = 0.095$), ($p = 0.171$). The linear regression model showed a very weak positive relationship, with SANU explaining only around 0.9% of the variance in PANU ($R^2 = 0.009$).

On the otherhand, PANU was significantly associated with EENU and the linear regression model showed a moderate positive association with EENU ($r = 0.474$), ($p < 0.001$). The PANU–EENU regression model explained around 22.5% of the variance in PANU ($R^2 = 0.225$).

These results suggest that elbow extension non-use (EENU) is more strongly associated with proximal arm non-use (PANU) than shoulder antepulsion non-use (SANU) in this dataset.

Several limitations should be considered including:

- Pearson's correlation and linear regression only assess linear relationships, so a possible non-linear association may not be detected.

- The scatterplots contain several observations far from the main cluster, and these potential outliers may influence both the correlation coefficients and regression models.
- The EENU model explains only 22.5% of the variance in PANU, indicating that EENU alone does not fully explain PANU. 77.5% is still left unexplained.
- Correlation and regression demonstrate association, not causation, so these results cannot show that EENU causes changes in PANU. People with greater EENU tend to also have greater PANU, which is an association, but the model can not say that a greater EENU causes greater PANU.
 - **Association:** two variables tend to change together.
 - **Causation:** changing one variable actually produces a change in the other.
- Rows containing missing values were removed from the analysis, which could introduce bias if the missing data were not random.

6. Statistical tests: what are statistical tests?

Reference

Vu, J., & Harrington, D. (2020). *Introductory statistics for the life and biomedical sciences* (1st ed.). OpenIntro. biostat statistics textbook

Test	Definition
Mean	The sum of all values divided by the total number of values. Ch. 1, 1.4.1 <i>Measures of center: mean and median</i> , p. 30
Median	The middle value when the observations are ordered from smallest to largest. Ch. 1, 1.4.1 <i>Measures of center: mean and median</i> , p. 30
Variance	A measure of how spread out the values are around the mean. A large variance means the values are more spread out. Ch. 1, 1.4.2 <i>Measures of spread</i> , p. 31
Standard deviation	Square root of the variance. Another measure of spread. How far values tend to be from the mean, in the same units as the data. Ch. 1, 1.4.2 <i>Measures of spread</i> , p. 31
Normal distribution	A symmetric, bell-shaped distribution where most values are close to the mean. A perfectly normal distribution, the mean, median, and mode are equal. Ch. 3, 3.3 <i>Normal distribution</i> , pp. 152–153
Statistical test	A method used to decide whether a difference or relationship in the sample is likely to be real or could just be due to random chance. usually starting with a null hypothesis (). used when we want to determine whether an observed difference/association in a sample provides sufficient evidence of a corresponding effect in the population, while accounting for sampling variability. Ch. 4, 4.3 <i>Hypothesis testing</i> , p. 212-214
Parametric test	A statistical test used for normally distributed data and makes more assumptions about the data. Assumptions: independent observations, a normal distribution and for some group comparisons, similar variances between groups. Ch. 4, 4.3 <i>Hypothesis testing</i> , p. 212-214 . Ch. 5, pp. 238, 245, 248

Test	Definition
Non-parametric test	A statistical test used for non-normally distributed data and makes fewer assumptions. Assumptions: Not normally distributed, appropriately measured data, independence of observations when comparing independent groups. Ch. 4, 4.3 <i>Hypothesis testing</i> , p. 212-214. Ch. 5, pp. 238, 245, 248
p-value	Tells us how unusual our result would be if there were actually no real effect or difference. A small p-value means the result would be unlikely under the null hypothesis. Ch. 4, 4.3.1, Step 4, p. 214
Paired test	Used when observations are related, such as measurements taken from the same participants before and after treatment. Ch. 5, 5.2 <i>Two-sample test for paired data</i> , pp. 244–246
Unpaired test	Used when the groups are independent, containing different observations or participants. Ch. 5, 5.3 <i>Two-sample test for independent data</i> , pp. 247–250

Risk of error when using a statistical test? (Ch. 4, 4.3.4 *Decision errors*, pp. 220–221)

- **Type I error** = You say there is an effect when there actually is not.
 - when we reject a true null hypothesis (false positive), with probability (α).
- **Type II error** = You say there is no effect when there actually is one.
 - when we fail to reject a false null hypothesis (false negative), with probability (β)

7. Effect of treatment over time

Q: Does a treatment training have an effect over time?

Treatment is a rehabilitation training for individuals with cardiac conditions.

a). Import data

```
# 1.Import the data

prepost <- read_csv("data/PrePost.csv")
```

```
Rows: 16 Columns: 2
— Column specification —————
Delimiter: ","
chr (1): time
dbl (1): perf

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# View the first 6 rows to make sure it was imported correctly
head(prepost)
```

```
# A tibble: 6 × 2
  perf time
<dbl> <chr>
```

```

1  125 Before
2  130 Before
3  132 Before
4  135 Before
5  136 Before
6  138 Before

```

b). Check the structure of the data

8 measurements before rehabilitation and 8 after:

```

Before: 125 130 132 135 136 138 140 145
After:  127 132 133 136 139 141 145 148

```

```

# Display the variable names
names(prepost)

```

```
[1] "perf" "time"
```

```

# Check the structure and data types
str(prepost)

```

```

spc_tbl_ [16 × 2] (S3: spec_tbl_df/tbl_df/tbl/data.frame)
 $ perf: num [1:16] 125 130 132 135 136 138 140 145 127 132 ...
 $ time: chr [1:16] "Before" "Before" "Before" "Before" ...
- attr(*, "spec")=
 .. cols(
 ..   perf = col_double(),
 ..   time = col_character()
 .. )
- attr(*, "problems")=<pointer: 0xbc08339b0>

```

```

# Obtain a first summary of the values
summary(prepost)

```

```

      perf      time
Min.   :125.0  Length :16
1st Qu.:132.0  N.unique : 2
Median :136.0  N.blank  : 0
Mean    :136.4  Min.nchar: 5
3rd Qu.:140.2  Max.nchar: 6
Max.    :148.0

```

c). Variables of the dataset

Variable	Description	Unit	Possible Values	Expected Change
perf	Performance measurement taken before and after cardiac rehabilitation	Not specified	Numerical values; observed in this dataset from 125 to 148	Expected to increase if a higher performance score represents better physical capacity

Variable	Description	Unit	Possible Values	Expected Change
time	Indicates whether the performance measurement was taken before or after rehabilitation	None	Before, After	Not applicable; this is the measurement occasion

d). Data analysis

Q: Does the rehabilitation treatment have an effect on measured performance from before to after treatment?

a) What comparison are we making?

- performance before rehabilitation vs performance after rehabilitation
 - Before: 125 130 132 135 136 138 140 145
 - After: 127 132 133 136 139 141 145 148
- If the same 8 participants are being compared before and after then this is a paired comparison

b) Parametric or non-parametric test?

- For a paired test, what matters is the distribution of the differences:
 - $difference = After - Before$
 - The differences are [2 2 1 1 3 3 5 3]
- Check that the differences are normally distributed using a **Shapiro-Wilk normality test**.
 - If $p > 0.05$ = not statistically significant and there is not enough evidence to say the differences depart from normality (we do not reject normality) → **parametric test** → **paired t-test**
 - If $p < 0.05$ = statistically significant and there is enough evidence to say the differences are not normally distributed (we reject normality) → **non-parametric test** → **Wilcoxon signed-rank test**.
 - For this dataset $p = 0.283 > 0.05$ therefore the data is normally distributed and a parametric paired t-test can be used for statistical analysis since the data are paired.
- The **paired t-test** gives $p = 0.001 < 0.05$ → there is a statistically significant difference between performance before and after rehabilitation.

```
# 1. Calculate the differences between before and after in the dataset
```

```
# a) Separate the before and after performance values
```

```
before <- prepost %>% filter(time == "Before") %>% pull(perf)
```

```
# only keep rows where time == "Before" and extract the perf column
```

```
after <- prepost %>% filter(time == "After") %>% pull(perf)
```

```
# only keep rows where time == "After" and extract the perf column
```

```
# b) Calculate the difference for each participant
```

```
difference <- after - before
```

```
difference
```

```
[1] 2 2 1 1 3 3 5 3
```

```
# 2. Check for normality of data using Shapiro-Wilk normality test
```

```
shapiro.test(difference)
```

Shapiro-Wilk normality test

```
data: difference
W = 0.89896, p-value = 0.2828
```

```
# 3. Paired t-test comparing performance before and after rehabilitation
t.test(after, before, paired = TRUE)
```

Paired t-test

```
data: after and before
t = 5.4006, df = 7, p-value = 0.001008
alternative hypothesis: true mean difference is not equal to 0
95 percent confidence interval:
 1.405392 3.594608
sample estimates:
mean difference
          2.5
```

c) Which graph illustrates the effect of rehabilitation treatment?

Since this is paired data, use a paired line plot to best show the effect of rehabilitation treatment on each participant.

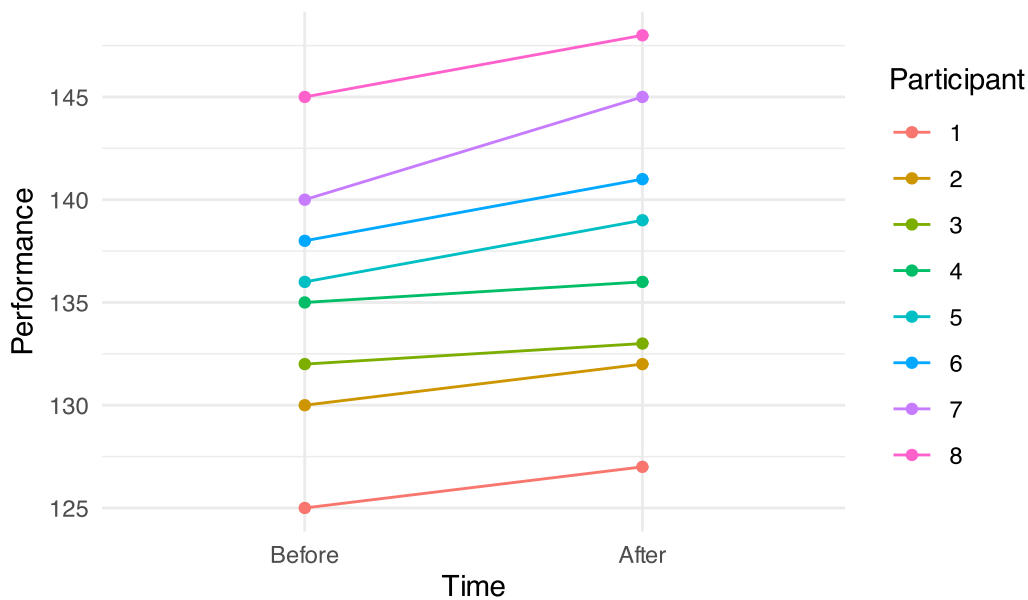
```
# 1. Reorganize data structure so each participant has a id (number 1-8)

# a). Add a new column to dataset: participant IDs to label each participant
prepost_id <- prepost %>% mutate(id = rep(1:8, 2), time = factor(time, levels = c("Before",
"After"))) # give participants a # 1-8 and repeat twice. Before comes first and after comes second.

# 2. Create the paired line plot
ggplot(prepost_id, aes(x = time, y = perf, group = id, color = factor(id))) +
  #color = factor(id) → converts numbers to categorical data to label each person with a colour
  # group = id → connect the correct person's points

  geom_line() +
  geom_point() +
  labs(title = "Performance Before and After Rehabilitation",
       x = "Time", y = "Performance", color = "Participant") + # create legend labeled participant
  theme_minimal()
```

Performance Before and After Rehabilitation



d) Rehabilitation treatment interpretation

Mean performance increased after rehabilitation compared with before. The slightly larger standard deviation after rehabilitation indicates that participant performance values were slightly more spread out around the mean.

- **Before rehabilitation:** 135.1 ± 6.2 = mean $\pm$ standard deviation
- **After rehabilitation:** 137.6 ± 7.0 = mean $\pm$ standard deviation
 - Smaller SD = participants values are more similar to each other
 - Larger SD = participant values are more spread out.

```
# 1. Calculate the mean & SD before rehabilitation
```

```
mean(before)
```

```
[1] 135.125
```

```
sd(before)
```

```
[1] 6.197638
```

```
# 2. Calculate the mean & SD after rehabilitation
```

```
mean(after)
```

```
[1] 137.625
```

```
sd(after)
```

```
[1] 7.008923
```

8. Testing Stereotypes

a). Import data

```
# 1.Import the data

snore <- read_delim("data/snore.csv") # delim separates the columns correctly

Rows: 100 Columns: 7
— Column specification —————
Delimiter: "\t"
chr (3): sex, snore, tabac
dbl (4): age, weight, height, alcool

i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
# View the first 6 rows to make sure it was imported correctly
head(snore)
```

```
# A tibble: 6 × 7
  age weight height alcool sex  snore tabac
<dbl> <dbl> <dbl> <dbl> <chr> <chr> <chr>
1    47     71   158     0 H      N      0
2    56     58   164     7 H      0      N
3    46    116   208     3 H      N      0
4    70     96   186     3 H      N      0
5    51     91   195     2 H      0      0
6    46     98   188     0 F      N      N
```

b). Data analysis

Q: Do the data confirm the following stereotypes?

a) Are snorers fatter?

The BMI of snorers and non-snores can be compared to gives us a measure of body mass relative to height, so it is more appropriate than simply asking whether snorers weigh more.

$BMI_{snorers}$ vs $BMI_{non-snorers}$

$$BMI = \frac{weight(kg)}{height(m)^2}$$

```
# 1. Create a new column BMI with the height m
snore <- snore %>% mutate(BMI = weight / (height / 100)^2) # mutate adds a new column BMI

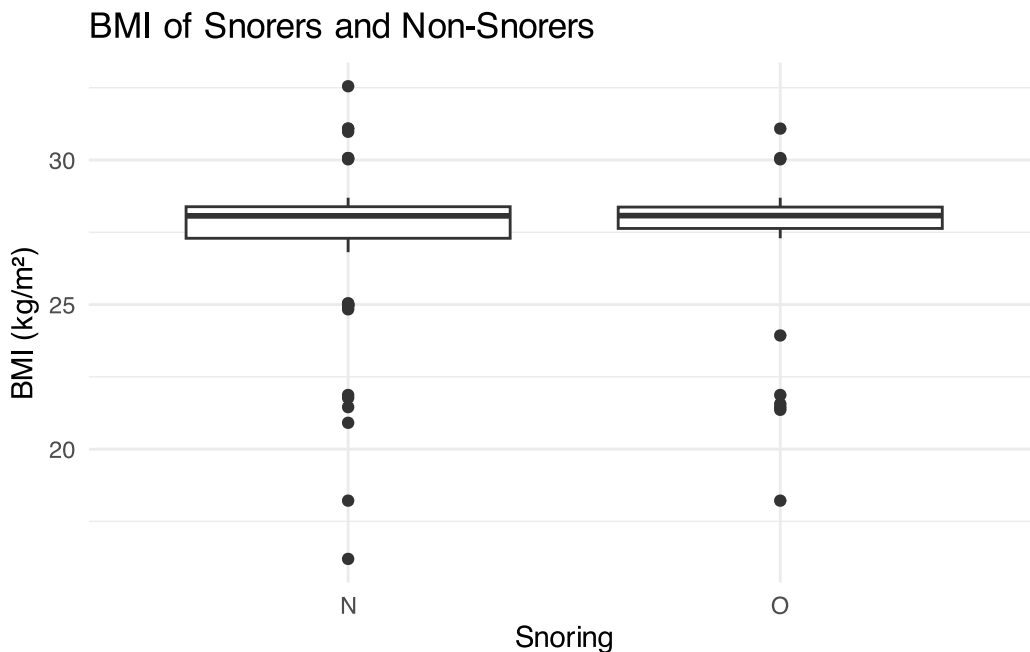
head(snore)
```

```
# A tibble: 6 × 8
  age weight height alcool sex  snore tabac  BMI
<dbl> <dbl> <dbl> <dbl> <chr> <chr> <chr> <dbl>
1    47     71   158     0 H      N      0   28.4
2    56     58   164     7 H      0      N   21.6
3    46    116   208     3 H      N      0   26.8
4    70     96   186     3 H      N      0   27.7
```

5	51	91	195	2 H	0	0	23.9
6	46	98	188	0 F	N	N	27.7

2. Make a boxplot to see the distribution of data before deciding the type of statistical test

```
ggplot(snore, aes(x = snore, y = BMI)) +
  geom_boxplot() +
  labs(title = "BMI of Snorers and Non-Snorers", x = "Snoring", y = "BMI (kg/m²)") +
  theme_minimal()
```



3. Create a table to look at the descriptive statistics

```
snore %>% group_by(snore) %>%
  summarise(mean_BMI = mean(BMI), median_BMI = median(BMI), sd_BMI = sd(BMI), n = n())
```

A tibble: 2 × 5

	snore	mean_BMI	median_BMI	sd_BMI	n
	<chr>	<dbl>	<dbl>	<dbl>	<int>
1	N	27.3	28.1	2.81	65
2	O	27.1	28.1	2.83	35

Both groups are extremely similar and the boxplot also shows a very similar distribution and spread in both groups. Therefore descriptively, there is no obvious evidence that snorers have a higher BMI.

- Non-snorers (N): mean BMI = 27.27, median = 28.07, SD = 2.81, (n=65)
- Snorers (O): mean BMI = 27.11, median = 28.08, SD = 2.83, (n=35)

These are **unpaired/independent groups**, because the snorers and non-snorers are different people.

Before deciding the type of statistical test: check to see if BMI is normally distributed in each group using the Shapiro-Wilk test.

4. Check the normality of BMI in both groups using Shapiro-Wilk test.

a) Check normality of BMI for non-snorers

```
shapiro.test(snore$BMI[snore$snore == "N"]) # take the BMI column only for the rows where snore
= no
```

Shapiro-Wilk normality test

```
data:  snore$BMI[snore$snore == "N"]
W = 0.78486, p-value = 2.323e-08
```

```
# b) Check normality of BMI for snorers
```

```
shapiro.test(snore$BMI[snore$snore == "0"]) # take the BMI column only for the rows where snore
= yes
```

Shapiro-Wilk normality test

```
data:  snore$BMI[snore$snore == "0"]
W = 0.72636, p-value = 9.81e-07
```

BMI is not normally distributed in either group. Both are much smaller than (0.05), so we reject the assumption of normality for both groups:

$$BMI_{snorers} \rightarrow p = 2.323e-08 < 0.05$$

$$BMI_{non-snorers} \rightarrow p = 9.81e-07 < 0.05$$

Since the two groups are independent and the data are non-normal, use the **Wilcoxon rank-sum test** (also called the **Mann-Whitney U test**). This tests whether the BMI distributions differ between snorers and non-snorers.

```
# 5. Run the Wilcoxon rank-sum test on BMI
```

```
wilcox.test(BMI ~ snore, data = snore)
```

Wilcoxon rank sum test with continuity correction

```
data:  BMI by snore
W = 1128.5, p-value = 0.951
alternative hypothesis: true location shift is not equal to 0
```

$p = 0.951 > 0.05 \rightarrow$ there is no statistically significant difference in BMI between snorers and non-snorers. The BMI distributions of snorers and non-snorers are not meaningfully different in this sample.

BMI was similar between snorers and non-snorers, with median values of 28.08 and 28.07 kg/m², respectively. The difference was not statistically significant (Wilcoxon rank-sum test, $W = 1128.5$, $p = 0.951$), providing no evidence to support the stereotype that snorers are fatter.

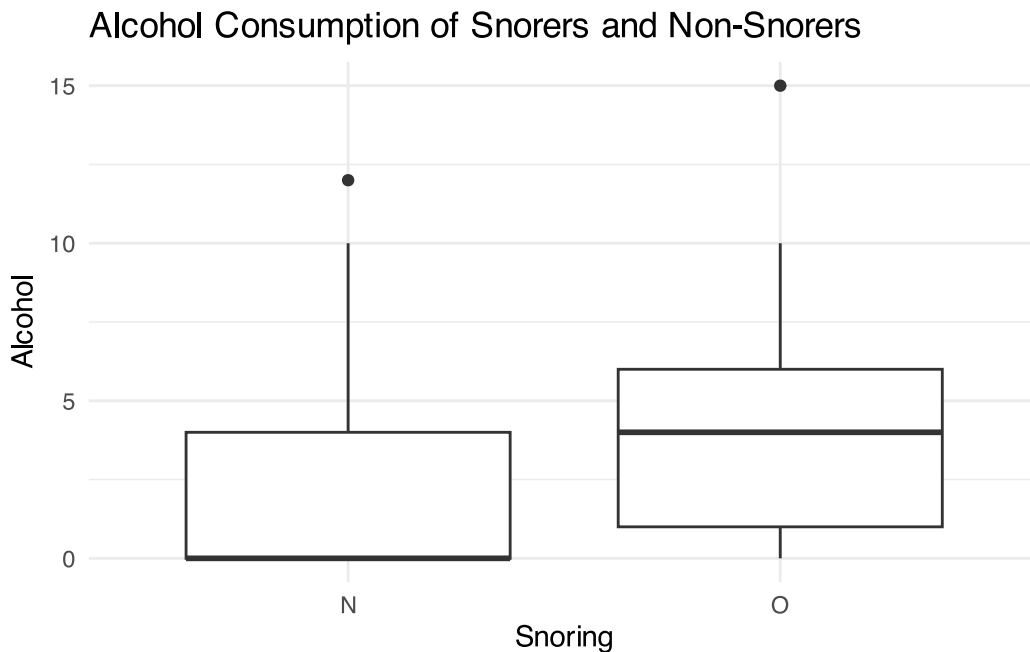
b) Do snorers drink or smoke more?

1. Do snorers drink more?

Alcohol consumption of snorers and non-snorers can be compared to determine whether snorers have higher alcohol consumption. Analyze the distribution of alcohol consumption before statistical analyses

$Alcohol_{snorers}$ vs $Alcohol_{non-snorers}$

```
# 1. Make a boxplot to compare alcohol consumption between snorers and non-snorers
ggplot(snore, aes(x = snore, y = alcool)) +
  geom_boxplot() +
  labs(title = "Alcohol Consumption of Snorers and Non-Snorers", x = "Snoring", y = "Alcohol") +
  theme_minimal()
```



```
# 2. Create a table to look at the descriptive statistics
```

```
snore %>%
  group_by(snore) %>%
  summarise(mean_alcool = mean(alcool), median_alcool = median(alcool), sd_alcool = sd(alcool),
    n = n())
```

```
# A tibble: 2 × 5
  snore mean_alcool median_alcool sd_alcool    n
  <chr>    <dbl>         <dbl>    <dbl> <int>
1 N         2.37           0      3.18    65
2 O         4.03           4      3.47    35
```

Snorers appear to consume more alcohol than non-snorers. Snorers had a higher mean alcohol consumption (4.03 vs 2.37) and a higher median (4 vs 0). The boxplot also shows generally higher alcohol values among snorers.

- Non-snorers (N): mean alcohol = 2.37, median = 0, SD = 3.18
- Snorers (O): mean alcohol = 4.03, median = 4, SD = 3.47

```
# 3. Check the normality of alcohol consumption in both groups using the Shapiro-Wilk test
```

```
# a) Check normality for non-snorers
shapiro.test(snore$alcool[snore$snore == "N"])
```

Shapiro-Wilk normality test

```
data: snore$alcohol[snore$snore == "N"]  
W = 0.76256, p-value = 6.818e-09
```

```
# b) Check normality for snorers  
shapiro.test(snore$alcohol[snore$snore == "0"])
```

Shapiro-Wilk normality test

```
data: snore$alcohol[snore$snore == "0"]  
W = 0.90387, p-value = 0.005032
```

Alcohol consumption was not normally distributed in either snorers or non-snorers (Shapiro–Wilk tests, both $p < 0.05$). Therefore, a non-parametric test was used to compare the two unpaired/independent groups.

$Alcohol_{snorers} \rightarrow p = 6.818e - 09 < 0.05$

$Alcohol_{non-snorers} \rightarrow p = 0.005032 < 0.05$

```
# 4. Run the Wilcoxon rank-sum test on alcohol consumption
```

```
wilcox.test(alcohol ~ snore, data = snore)
```

Wilcoxon rank sum test with continuity correction

```
data: alcohol by snore  
W = 788, p-value = 0.008594  
alternative hypothesis: true location shift is not equal to 0
```

This result is statistically significant, meaning there is a difference between the two groups.

$p = 0.008594 < 0.05$

Snorers had higher alcohol consumption than non-snorers, with median values of 4 and 0, respectively. This difference was statistically significant (Wilcoxon rank-sum test, $W = 788$, $p = 0.009$), supporting the stereotype that snorers drink more.

2. Do snorers smoke more?

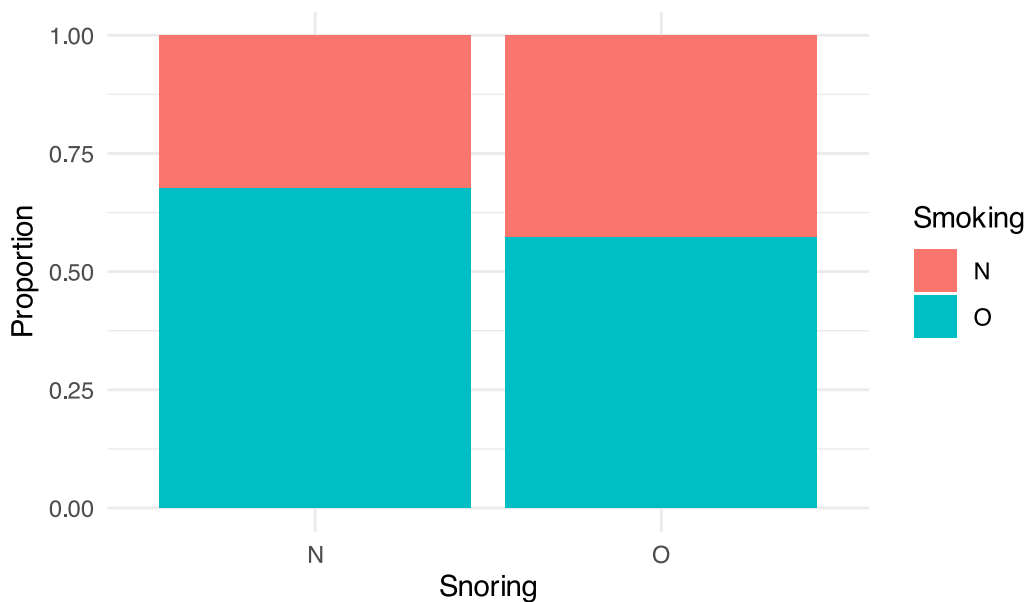
Alcohol consumption of smokers and non-smokers can be compared to determine whether smokers have higher alcohol consumption. Analyze the proportion of smokers between snorers and non-snorers and since smoking and alcohol is categorical (0/N) data, therefore a histogram was plotted rather than a box-pot.

$Smoker_{non-snorers}$ vs $Smoker_{snorers}$

```
# 1. Compare smoking distribution between snorers and non-snorers
```

```
ggplot(snore, aes(x = snore, fill = tabac)) +  
  geom_bar(position = "fill") +  
  labs(title = "Smoking Status of Snorers and Non-Snorers", x = "Snoring", y = "Proportion",  
fill = "Smoking") +  
  theme_minimal()
```

Smoking Status of Snorers and Non-Snorers



```
# 2. Create a contingency table
```

```
smoke_table <- table(snore$snore, snore$tabac)
```

```
# a). Calculate proportions within each snoring group
```

```
prop.table(smoke_table, margin = 1) # Calculate the percentages within each row, each row adds up to 100%
```

```
      N      O
N 0.3230769 0.6769231
O 0.4285714 0.5714286
```

```
# b). Convert the proportions to %
```

```
prop.table(smoke_table, margin = 1) * 100
```

```
      N      O
N 32.30769 67.69231
O 42.85714 57.14286
```

The descriptive statistics suggest that snorers do not smoke more in this sample. About 57.1% of snorers were smokers compared with 67.7% of non-snorers, so the proportion of smokers was actually lower among snorers.

- Among non-snorers, 32.31% are non-smokers and 67.69% are smokers
- Among snorers, 42.86% are non-smokers and 57.14% are smokers

Because both variables are categorical data (0/N), a **Chi-square test of independence** instead of a parametric or non-parametric test since the data is categorical and both groups are independent:

```
# 3. Test whether smoking status is associated with snoring status
```

```
chisq.test(smoke_table)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: smoke_table
```

```
X-squared = 0.68872, df = 1, p-value = 0.4066
```

No statistically significant association between smoking status and snoring status. Descriptively, snorers actually smoked less often in this sample (57.1% vs 67.7%), but the Chi-square test shows that this difference is not statistically significant ($\chi^2(1) = 0.689, p = 0.407$), so the data do not support the stereotype that snorers smoke more.

$p = 0.407 > 0.05$

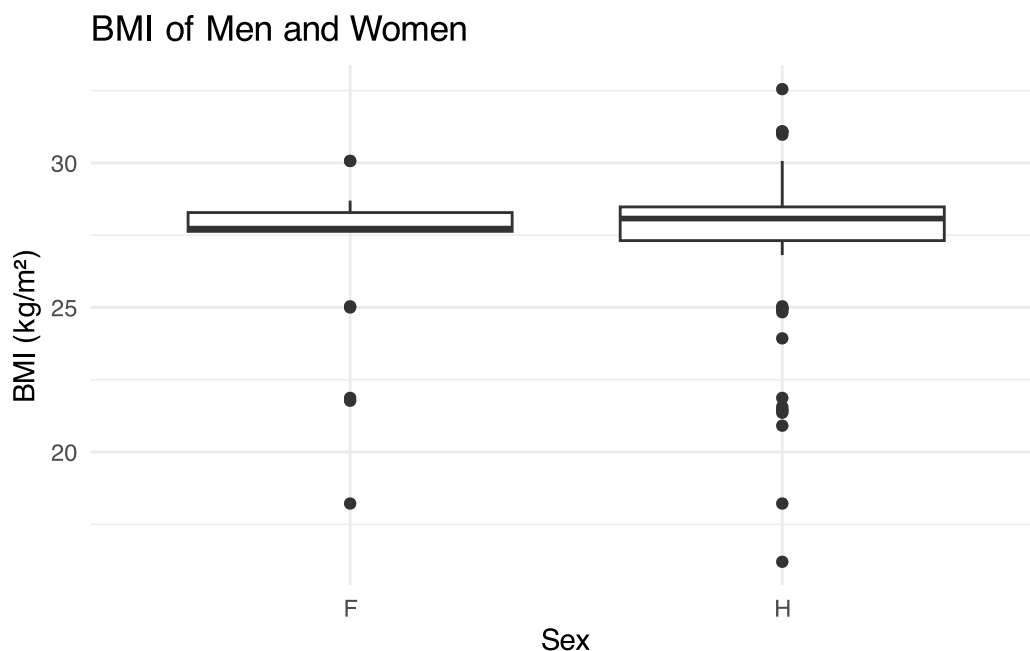
c) Are men fatter?

We can compare BMI between men and women. The distribution of BMI between men and women are first analysed using a box-plot before statistical analyses is performed.

BMI_{men} vs BMI_{women}

```
# 1. Make a boxplot to compare BMI between men and women
```

```
ggplot(snore, aes(x = sex, y = BMI)) +  
  geom_boxplot() +  
  labs(title = "BMI of Men and Women", x = "Sex", y = "BMI (kg/m²)") +  
  theme_minimal()
```



```
# 2. Create a table to look at the descriptive statistics
```

```
snore %>%  
  group_by(sex) %>%  
  summarise(mean_BMI = mean(BMI), median_BMI = median(BMI), sd_BMI = sd(BMI), n = n())
```

```
# A tibble: 2 × 5
  sex    mean_BMI median_BMI sd_BMI    n
<chr>    <dbl>      <dbl>  <dbl> <int>
1 F        27.0        27.7   2.72    25
2 H        27.3        28.1   2.85    75
```

Descriptive results suggest that men have a slightly higher BMI than women, but the difference looks small. The median BMI was 28.08 kg/m² for men and 27.73 kg/m² for women, while the mean BMI was 27.27 and 27.04 kg/m², respectively.

- Women (F): mean BMI = **27.04**, median = **27.73**, SD = **2.72**, (n=25)
- Men (H): mean BMI = **27.27**, median = **28.08**, SD = **2.85**, (n=75)

Check the normality of the data using the Shapiro-Wilk test

```
# 3. Check normality of BMI in both groups using the Shapiro-Wilk test
```

```
# a) Women
```

```
shapiro.test(snore$BMI[snore$sex == "F"])
```

Shapiro-Wilk normality test

```
data:  snore$BMI[snore$sex == "F"]
```

```
W = 0.71462, p-value = 1.154e-05
```

```
# b) Men
```

```
shapiro.test(snore$BMI[snore$sex == "H"])
```

Shapiro-Wilk normality test

```
data:  snore$BMI[snore$sex == "H"]
```

```
W = 0.78184, p-value = 3.456e-09
```

BMI was not normally distributed in either women or men (Shapiro–Wilk tests, both $p < 0.05$). Therefore, a non-parametric test was used to compare BMI between the two independent groups.

- Men: $p = 1.154e - 05 < 0.05$
- Women: $p = 3.456e - 09 < 0.05$

```
# 4. Run the Wilcoxon rank-sum test on BMI
```

```
wilcox.test(BMI ~ sex, data = snore)
```

Wilcoxon rank sum test with continuity correction

```
data:  BMI by sex
```

```
W = 850, p-value = 0.4884
```

```
alternative hypothesis: true location shift is not equal to 0
```

No statistically significant difference in BMI between men and women.

- $p = 0.488 > 0.05$

Men had a slightly higher median BMI than women (28.08 vs 27.73 kg/m²), but the difference was not statistically significant (Wilcoxon rank-sum test, $W = 850$, $p = 0.488$). Therefore, the data do not support the stereotype that men are fatter.

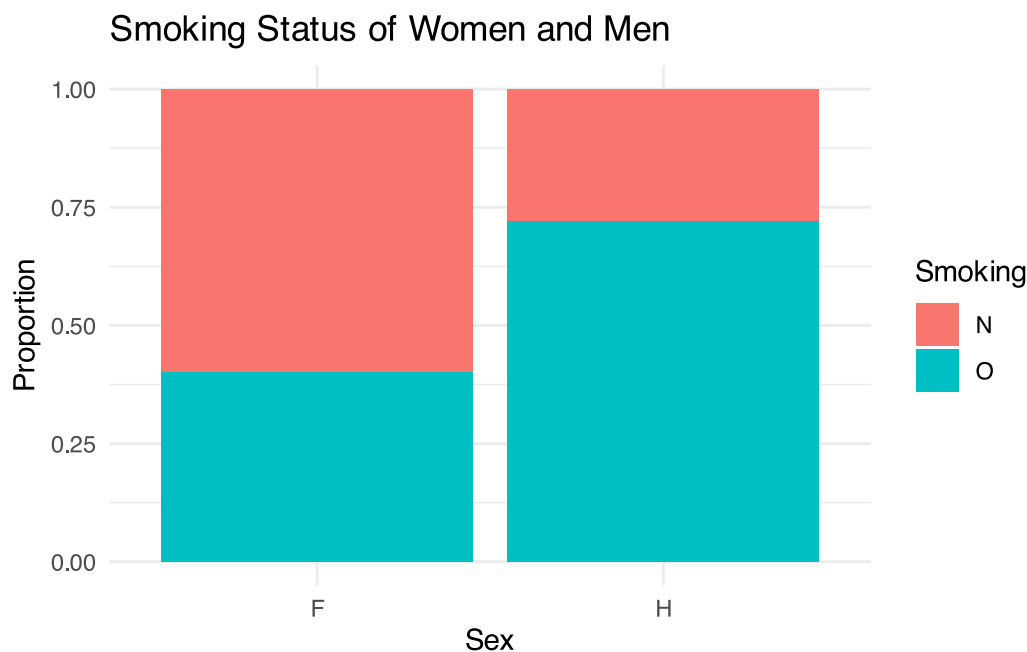
d) Do women smoke less?

Both gender and `smoking` are categorical variables therefore the proportion of smokers among women and men must be compared.

$Smokers_{men}$ vs $Smokers_{women}$

```
# 1. Make a proportional bar plot to compare smoking status between women and men
```

```
ggplot(snore, aes(x = sex, fill = tabac)) +  
  geom_bar(position = "fill") +  
  labs(title = "Smoking Status of Women and Men", x = "Sex", y = "Proportion",  
        fill = "Smoking") +  
  theme_minimal()
```



```
# 2. Create a contingency table for sex and smoking status
```

```
sex_smoke_table <- table(snore$sex, smoke$tabac)
```

```
# a) Calculate proportions within each sex
```

```
prop.table(sex_smoke_table, margin = 1)
```

```
      N      O  
F 0.60 0.40  
H 0.28 0.72
```

```
# b) Convert the proportions to percentages
```

```
prop.table(sex_smoke_table, margin = 1) * 100
```

	N	0
F	60	40
H	28	72

A lower proportion of women were smokers compared with men (40% vs 72%). Descriptively, this supports the stereotype that women smoke less.

- **Women (F):** 40% smokers, 60% non-smokers
- **Men (H):** 72% smokers, 28% non-smokers

Since both gender and smoking are categorical variables a Chi-square test is performed

```
# 3. Test whether smoking status is associated with sex

chisq.test(sex_smoke_table)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: sex_smoke_table
X-squared = 7.0023, df = 1, p-value = 0.00814
```

There is significant association between sex and smoking status.

- $p = 0.008 < 0.05$

A lower proportion of women were smokers compared with men (40% vs 72%). This difference was statistically significant ($\chi^2(1) = 7.00, p = 0.008$), supporting the stereotype that women smoke less in this sample.

e) From a more general perspective: are there any correlations between variables?

We are no longer comparing two groups. We want to see whether the numerical variables tend to change together.

- **Numerical variables:** age, weight, height, alcohol
- **Categorical variables:** snore, gender, smoking

Categorical data should not be put in a standard correlation matrix since they are not numerical quantities. Based on the previous analysis above, some numerical variables were not normally distributed, therefore **Spearman correlation** can be used to see whether two variables generally increase or decrease together. Spearman is used instead of Pearson because Spearman does not require normality.

1. A dataset must be created for only the numerical variables
2. The Spearman correlation matrix is calculated
3. The interesting correlations are then tested to see if they are statistically significant or not

```
# 1. Select the numerical variables for the correlation analysis

snore_numeric <- snore %>%
  select(age, weight, height, alcohol)

head(snore_numeric)
```

```
# A tibble: 6 × 4
  age weight height alcohol
<dbl> <dbl> <dbl> <dbl>
1    47     71    158      0
```

2	56	58	164	7
3	46	116	208	3
4	70	96	186	3
5	51	91	195	2
6	46	98	188	0

```
# 2. Calculate Spearman correlations between all numerical variables

correlation_matrix <- cor(snore_numeric, method = "spearman", use = "complete.obs")

correlation_matrix
```

```
          age      weight      height      alcool
age      1.0000000000 -0.01455125 0.008072461 -0.10748796
weight -0.014551254  1.000000000 0.930383489  0.08568687
height  0.008072461  0.93038349 1.000000000  0.12539769
alcool -0.107487956  0.08568687 0.125397694  1.00000000
```

The main relationship is between weight and height. It is a very strong positive correlation. Taller participants generally tend to weigh more. All other correlations were weak or close to zero.

- $r_s = 0.930$
- 1.0 = variables are perfectly correlated with each other

Now the correlation between weight and height can be tested to see whether it is statistically significant or not. **Spearman correlation** is used to test the significance. Spearman correlation was used because some numerical variables were not normally distributed as mentioned previously, making a non-parametric correlation more appropriate.

```
# 3. Test the correlation between height and weight

cor.test(snore$height, snore$weight, method = "spearman")
```

```
Warning in cor.test.default(snore$height, snore$weight, method = "spearman"):
cannot compute exact p-value with ties
```

Spearman's rank correlation rho

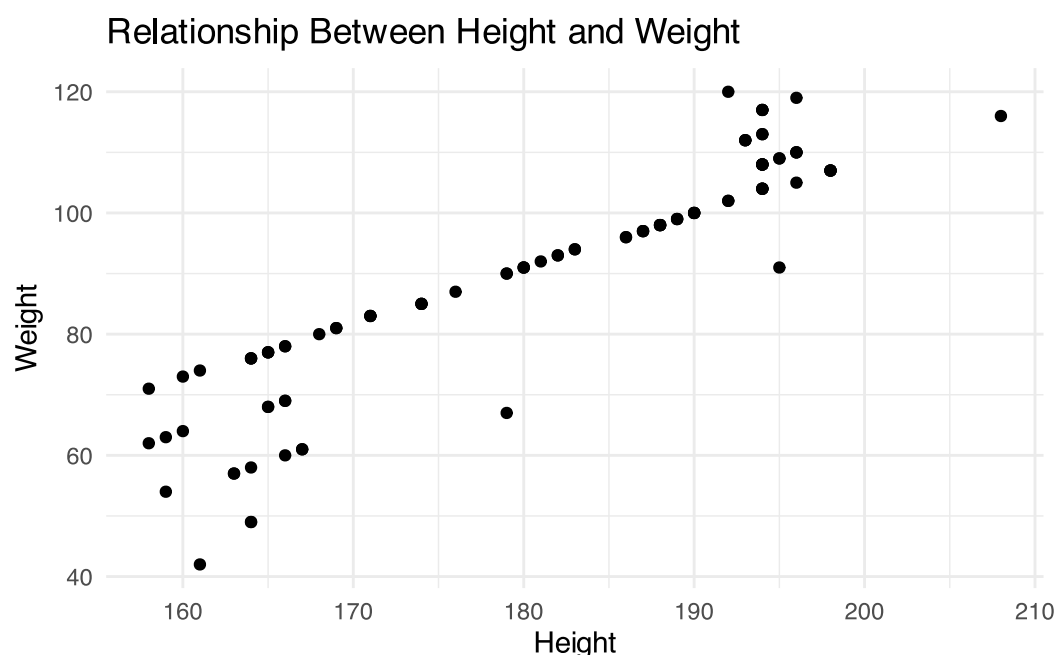
```
data: snore$height and snore$weight
S = 11602, p-value < 2.2e-16
alternative hypothesis: true rho is not equal to 0
sample estimates:
      rho
0.9303835
```

There is a very strong positive and statistically significant association between height and weight: as height increases, weight generally increases as well. $r_s = 0.930$

- $p < 2.2e - 16$

```
# 4. Visualize the relationship between height and weight
```

```
ggplot(snore, aes(x = height, y = weight)) +
  geom_point() +
  labs(title = "Relationship Between Height and Weight", x = "Height", y = "Weight") +
  theme_minimal()
```



Height and weight showed a very strong positive correlation ($r_s = 0.930$, $p = 2.2e - 16$), indicating that taller participants generally had greater body weight. The other numerical variables showed only weak or negligible correlations.

9. Beyond the scope

The statistical methods used in this assignment included:

Methods	**Description*
Descriptive statistics	Summarized the data using mean, median, SD, percent-ages, etc.
Boxplots / scatterplots	Visualized distributions and relationships
Paired t-test	Compared two related measurements, such as before and after, when the differences were approximately normally distributed
Wilcoxon rank-sum test	Compared two independent groups when data were non-normal
Chi-square test	Tested relationships between categorical variables
Spearman correlation	Tested whether two numerical variables were related when the data were non-normally distributed

The more advanced methods mentioned in the epilogue are different because they can work with several variables at the same time:

More advanced methods	What they could do
Linear regression	Use several variables to predict a numerical value
Logistic regression	Use several variables to predict a category/probability, such as snorer vs non-snorer
Clustering	Automatically find groups of similar individuals

More advanced methods	What they could do
PCA	Combine many related variables into fewer variables to simplify the dataset
• So far, the analysis has mainly used basic statistical methods to compare variables or study relationships between them, such as t-tests, Wilcoxon tests, Chi-square tests and correlations. • More advanced data-mining methods could go further by analysing several variables at the same time. For example, age, weight, height, alcohol consumption, sex and smoking status could potentially be combined using logistic regression to predict whether a person snores. Linear regression could be used when the variable being predicted is numerical instead of categorical. • Other methods can be used for different purposes. Clustering could identify groups of people with similar characteristics without defining the groups beforehand, while PCA could simplify a dataset containing many related variables by reducing them to a smaller number of combined variables. • These methods could reveal patterns that are difficult to see when variables are studied separately. However, the research question should always come first, because finding a statistical pattern does not automatically mean that the pattern is useful or meaningful.	